Type MS · Use a spirometer and identify lung volumes on its trace

Answer 1 of 8 · Question 5 part b.i · 4 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

<table>
<tr><td>5 (a) (i)</td><td>any four of

• exhale through tube A;
• time how long for limewater to go cloudy;
• replace limewater;
• inhale through tube B;
• compare time to go cloudy;</td><td></td><td>4 marks</td></tr>
<tr><td>(ii)</td><td>• more carbon dioxide in exhaled air;
• quicker/less time to go cloudy with exhaled air;
• produced by respiration;</td><td></td><td>3 marks</td></tr>
<tr><td>(b) (i)</td><td><table><tr><th>Lung volume</th><th>Letter</th><th>Volume in dm^3</th></tr><tr><td>tidal volume</td><td>W;</td><td>0.4;</td></tr><tr><td>vital capacity</td><td>Z;</td><td>2.25;</td></tr></table></td><td></td><td>4 marks</td></tr>
<tr><td>(ii)</td><td>• deep/maximum inhale;
• followed by maximum/forcefully exhale;</td><td>must be in correct order</td><td>2 marks</td></tr>
</table>

Total 13 marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 1 of 8 · Question 5 part b.i · 4 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 2 of 8 · Question 6 part b.i · 4 marks

2020 JAN · Paper 01 · find part b.i in this mark scheme extract

Question number				
6	(a)	bronchus/bronchi (1)		1
	(b) (i)	• inhale and exhale/breathe in an out (1) • through mouthpiece (1) • floating air chamber rises/falls (1) • pen records a trace on (the rotating) graph paper/drum (1)		4
	(ii)	Do not share the mouthpiece with another person/disinfect mouthpiece before use by another person/after use (1)		1
	(iii)	Any two from: • airways /bronchi/bronchus constricted/obstructed/reduced diameter (1) • less/reduced airflow (1) • per unit time (1)		2
	(c) (i)	cannot be transmitted from one person to another/not contagious/not caused by a pathogen/bacteria/virus/microorganism (1)		1
	(ii)	9.5 - 7 = 2.5(1) 2.5 ÷ 7 x 100(1) 36(%) (1)	Full marks for correct final answer Allow 35.7 for final answer	3
	(iii)	people taking less exercise/ consuming too much fat/sugar/ taking in more energy than is being used/ increase in the number of people smoking/drinking more alcohol/obesity (1)	Allow other valid lifestyle factor	1

Total for Question 6 = 13 marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 3 of 8 · Question 2 part c · 2 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

number				
2	(a)	• oxygen; • glucose; • water; • carbon dioxide;	in any order in any order	2 2
	(b) (i)	• A to B = 60 seconds; • 10 breaths during that period = 10 per minute;	Allow 60 anywhere in response	2
	(ii)	• 19.5 - 13.5 squares of movement = 6 squares; • 1dm^3 = 4 squares; • 6/4 = 1.5dm^3;		3
	(c)	• peaks/troughs higher/lower; • closer together;	Allow increase in amplitude Allow increase in frequency Answers must relate to peaks/waves/curves Ignore details relating to scientific details such as number or depth of breaths.	2

Total 11 marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 3 of 8 · Question 2 part c · 2 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 4 of 8 · Question 5 part b.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

<table>
<tr><td>5 (a) (i)</td><td>any four of

• exhale through tube A;
• time how long for limewater to go cloudy;
• replace limewater;
• inhale through tube B;
• compare time to go cloudy;</td><td></td><td>4 marks</td></tr>
<tr><td>(ii)</td><td>• more carbon dioxide in exhaled air;
• quicker/less time to go cloudy with exhaled air;
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• followed by maximum/forcefully exhale;</td><td>must be in correct order</td><td>2 marks</td></tr>
</table>

Total 13 marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 4 of 8 · Question 5 part b.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 5 of 8 · Question 7 part c.i · 1 mark

2024 JUN · Paper 01 · find part c.i in this mark scheme extract

7 (a)		• intercostal muscles contract; • ribs move up and out; • diaphragm contracts and flattens; • air is drawn into the lungs;	ignore ref to internal/external	(4)
(b)		• detect pH of blood/levels of CO_2; • to maintain homeostasis/regulate breathing/heart rate;		(2)
(c)	(i)	tidal (volume);		(1)
	(ii)	residual (volume);		(1)
	(iii)	vital capacity;		(1)

Total for Question 7 = 9 Marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 6 of 8 · Question 7 part c.ii · 1 mark

2024 JUN · Paper 01 · find part c.ii in this mark scheme extract

7 (a)		• intercostal muscles contract; • ribs move up and out; • diaphragm contracts and flattens; • air is drawn into the lungs;	ignore ref to internal/external	(4)
(b)		• detect pH of blood/levels of CO_2; • to maintain homeostasis/regulate breathing/heart rate;		(2)
(c)	(i)	tidal (volume);		(1)
	(ii)	residual (volume);		(1)
	(iii)	vital capacity;		(1)

Total for Question 7 = 9 Marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 7 of 8 · Question 7 part c.iii · 1 mark

2024 JUN · Paper 01 · find part c.iii in this mark scheme extract

7 (a)		• intercostal muscles contract; • ribs move up and out; • diaphragm contracts and flattens; • air is drawn into the lungs;	ignore ref to internal/external	(4)
(b)		• detect pH of blood/levels of CO_2; • to maintain homeostasis/regulate breathing/heart rate;		(2)
(c)	(i)	tidal (volume);		(1)
	(ii)	residual (volume);		(1)
	(iii)	vital capacity;		(1)

Total for Question 7 = 9 Marks

Type MS · Use a spirometer and identify lung volumes on its trace

Answer 8 of 8 · Question 4 part b.i · 4 marks

2024 JUN · Paper 02 · find part b.i in this mark scheme extract

Number				
4 (a)	(i)	Correct labels to: trachea; bronchus/bronchi; alveoli;		(3)
	(ii)	• many alveoli/large surface area; • short diffusion distance/ alveoli 1 cell thick/capillary 1 cell thick/ thin wall; • moist; • (dense/extensive/large) network of capillaries;		(4)
(b)	(i)	any four from • reference to use of a nose clip; • reference to soda lime (to absorb carbon dioxide) • reference to sterilising mouth piece; • breathe in and out(normally) through a mouthpiece(connected to spirometer); • pen goes up and down (on a rotating drum);\ • measure peak to trough/ highest to lowest/top to bottom/ from the trace;		(4)
	(ii)	4600 ÷ 30 (x 100) = 1380; 4600 – 1380 = 3220 (cm^3); or 4600×0.7=3220 (cm^3);;	ecf: allow correct calculation from values given for mp1. Max 2 marks	(2)

Total for Question 4 = 13 marks